

Axon: A Standardized Open-Source Pipeline for Brain Organoid MEA Electrophysiology — Validation Across Four Public Datasets

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github.com/metin/axon

Abstract

Brain organoid MEA (microelectrode array) electrophysiology is a rapidly growing field with no established analysis standards. Different groups use different metrics, thresholds, and algorithms — making cross-study comparison difficult. We present **Axon** (v6.3), an open-source Python pipeline for end-to-end analysis of NWB-formatted brain organoid MEA recordings with direct DANDI Archive integration. Axon computes per-unit firing rate, inter-spike interval statistics, coefficient of variation, Spike Time Tiling Coefficient (STTC) cross-unit synchrony matrices, adaptive network burst detection, and graph-theoretic network topology via a single command-line interface. Version 6.3 adds full **SpikeInterface** integration for spike sorting of raw high-density MEA data, supporting four algorithms (mountainsort5, spykingcircus2, tridesclous2, simple) with DC-corrected RMS-based channel selection (default 64 of 942 channels) and per-unit quality metrics (SNR, ISI violation ratio, firing rate). We validate the pipeline on four public DANDI datasets spanning **29 subjects**: DANDI:001603 (human and mouse brain organoids and ex vivo mouse cortex, sorted units), DANDI:001132 (human hippocampal organoids, HD-MEA raw signal), DANDI:000774 (in vitro MEA, sorted units), and DANDI:001611 (chronic dissociated cortical culture, scalability test). All analysis modules are dataset-agnostic. Source code is available at github.com/metin/axon.

Keywords: brain organoid, MEA electrophysiology, NWB, DANDI, spike sorting, SpikeInterface, STTC, graph theory, open-source, organoid intelligence

1. Introduction

Brain organoids derived from induced pluripotent stem cells (iPSCs) have emerged as a tractable three-dimensional model for human cortical development [Lancaster et al., 2013], disease modeling [Trujillo and Bhanu, 2022], and — most recently — as a substrate for biological computation [Kagan et al., 2022]. Coupled to microelectrode arrays (MEAs), organoids generate measurable extracellular electrical activity, including spontaneous firing, network bursts, and coordinated oscillations [Trujillo et al., 2019].

Despite rapid growth, no standardized analysis pipeline exists for organoid MEA electrophysiology data. Different groups use different spike detection thresholds, burst detection algorithms, and synchrony metrics — making cross-study comparison difficult [Smirnova et al., 2023]. The Neurodata Without Borders (NWB) format [Rubel et al., 2022] and the DANDI Archive [Bhatt et al., 2023] provide a common data standard, but the analytical layer remains fragmented.

The availability of public organoid MEA datasets — particularly DANDI:001603 [van der Molen et al., 2026] — creates an opportunity to develop and validate a standardized pipeline across multiple preparations, species, and recording conditions. However, these datasets contain both pre-sorted units (NWB units table) and raw high-density MEA recordings (ElectricalSeries only), requiring a pipeline that handles both data types.

We present **Axon** (v6.3), a modular Python pipeline that addresses this gap. Axon processes NWB files from the DANDI Archive through a standardized analysis chain: quality control, per-unit ISI statistics, STTC synchrony matrices, adaptive burst detection, and graph-theoretic network topology. For raw signal files, Axon integrates SpikeInterface [Buccino et al., 2020] for automated spike sorting with four algorithm options, DC-corrected RMS-based channel selection, and quality metrics per sorted unit. All modules are dataset-agnostic.

We validate the pipeline on 29 subjects from four public DANDI datasets, spanning human and mouse brain organoids, ex vivo cortical slices, high-density hippocampal organoid recordings, and chronic dissociated cortical cultures. We report cross-dataset electrophysiological profiles, age-dependent synchrony in human organoids, network topology transitions, and SpikeInterface performance on HD-MEA data.

2. Methods

2.1. Pipeline Architecture

Axon is implemented in Python 3.9+ and organized into 15 modules (Table 1). The pipeline detects NWB file type automatically: if a units table is present, per-unit analysis proceeds directly; if only raw ElectricalSeries data is present, the user is offered threshold detection or SpikeInterface spike sorting. All outputs are written to a subject-specific results directory as CSV files and a 4-panel quicklook figure.

Table 1: Axon v6.3 core modules.

Module	Function
organoid_io.py	NWB reading, file type detection (sorted/raw), RAM safety, DANDI streaming
organoid_signal.py	Bandpass filter (300–3000 Hz), MAD-threshold spike detection
organoid_qc.py	Refractory violation rate, Wilson CI, SNR
organoid_metrics.py	ISI statistics, CV, STTC ($\Delta t = 10$ ms), permutation test
organoid_units_analiz.py	Per-unit orchestrator, STTC matrix, burst detection, quicklook
si_sorting.py	SpikeInterface: 4 algorithms, RMS channel selection, quality metrics
graph_analiz.py	Clustering coefficient, path length, small-world index σ
gruplu_analiz.py	Auto-grouping by prefix/duration, Mann–Whitney U
organoid_cli.py	CLI: search, download, analyze in one command

2.2. Key Algorithms

2.2.1. Spike Time Tiling Coefficient (STTC)

Cross-unit pairwise synchrony was quantified using the STTC metric [Cutts and Eglen, 2014], which is independent of firing rate. For two spike trains A and B with coincidence window $\Delta t = 10$ ms:

$$\text{STTC} = \frac{1}{2} \left[\frac{P_A - T_B}{1 - P_A \cdot T_B} + \frac{P_B - T_A}{1 - P_B \cdot T_A} \right] \quad (1)$$

where T_A is the fraction of time within Δt of a spike in A , and P_A is the fraction of B spikes within Δt of an A spike. STTC ranges from -1 (anti-correlated) to $+1$ (perfectly synchronous). Matrices were computed for the top 20 most active units per subject.

2.2.2. Adaptive Burst Detection

Population firing rate was binned at 50 ms. Burst threshold was set adaptively using the Median Absolute Deviation (MAD):

$$\theta = \text{median}(r) + 3 \times \text{MAD}(r), \quad \text{with } \geq 3 \text{ co-firing units required} \quad (2)$$

This adaptive approach avoids the fixed-threshold problem when comparing preparations with 10-fold differences in baseline firing rate.

2.2.3. Graph-Theoretic Network Topology

STTC values exceeding 0.1 defined network edges. Clustering coefficient C , mean path length L , and small-world index $\sigma = (C/C_{\text{rand}})/(L/L_{\text{rand}})$ were computed via NetworkX. $\sigma > 1$ indicates small-world topology.

2.3. SpikeInterface Integration

For raw electrophysiology data, Axon integrates SpikeInterface (v0.104.3) [Buccino et al., 2020] via the `si_sorting.py` module. The workflow:

1. NWB reading with automatic ElectricalSeries path detection
2. Unsigned-to-signed dtype conversion (required for uint16 HD-MEA data)
3. **DC-offset removal** followed by per-channel RMS computation (10-second window)
4. Selection of N most active channels by RMS (default $N = 64$)
5. Bandpass filtering (300–3000 Hz)
6. Spike sorting: mountainsort5 (scheme1), spykingcircus2, tridesclous2, or simple
7. Quality metrics: SNR, ISI violation ratio (threshold $< 10\%$), firing rate (threshold > 0.05 Hz)

DC-offset correction is critical for uint16 recordings: without it, all channels exhibit near-identical RMS ($\approx 32,000$ ADU), rendering channel selection meaningless. After mean subtraction, RMS reflects genuine signal amplitude variation (13–68 μV across channels in DANDI:001132), enabling meaningful selection of the most active electrode subset.

2.4. Datasets

Four public DANDI datasets were analyzed (Table 2). All data were accessed under CC-BY-4.0 licenses. No new biological experiments were conducted.

3. Results

3.1. Pipeline Validation — Synthetic Tests

Axon passed 15/15 synthetic validation tests covering: spike detection accuracy; STTC correctness on known inputs (identical trains: STTC = 1.000; independent Poisson trains: STTC = -0.015);

Table 2: Datasets analyzed in this study.

Dataset	Description	Subjects	Data type	Species
DANDI:001603 [van der Molen et al., 2026]	Preconfigured firing sequences, brain organoids	18	Sorted units	Human, mouse
DANDI:001132	Multimodal HD-MEA, hippocampal organoids	5	Raw signal	Human
DANDI:000774	Spatial/temporal propagation, in vitro MEA	5	Sorted units	Mouse
DANDI:001611	Chronic dissociated cortical cultures	1	Sorted units	Rat

MAD robustness (5% outlier injection: std +403%, MAD +0.8%); RAM overflow protection; and Wilson CI correctness at zero refractory violations.

3.2. DANDI:001603 — Cross-Category Electrophysiological Profiles

The complete DANDI:001603 dataset (18 subjects with sorted units) was analyzed. Three biological categories were identified from subject metadata (Table 3).

Table 3: DANDI:001603 three-category descriptive statistics (mean $\pm$ SD). Recording durations and ages were not controlled across categories.

Category	n	Median Hz	Median CV	Median STTC	Burst/min	Active
Human organoid	8	0.42 ± 0.18	1.92 ± 0.85	0.159 ± 0.150	9.84 ± 6.18	99%
Mouse organoid	5	4.42 ± 1.48	1.15 ± 1.04	0.014 ± 0.030	5.40 ± 10.80	49%
Ex vivo mouse cortex	5	0.39 ± 0.18	2.83 ± 1.04	0.144 ± 0.060	10.10 ± 5.43	100%

Mouse organoids showed approximately 10-fold higher median firing rate than human organoids (4.42 vs. 0.42 Hz). Ex vivo mouse cortex showed a firing rate profile comparable to human organoids (0.39 Hz) despite the same species origin, suggesting that the firing rate difference reflects preparation-specific factors rather than species genetics alone. Age differences, culture protocols, and MEA geometry were not controlled and remain as confounds for future investigation.

3.3. Age-Dependent Synchrony in Human Organoids

Human organoids were divided into two age groups based on available metadata: G1 (6–7 months, HO1–HO4, $n = 4$) and G2 (≈ 3.3 months/P100D, HO5–HO8, $n = 4$). Mann–Whitney U test revealed significant differences in both median STTC (0.295 ± 0.096 vs. 0.023 ± 0.008 , $p = 0.029$) and median CV (1.17 ± 0.16 vs. 2.68 ± 0.53 , $p = 0.029$). Burst rate and median Hz did not differ significantly ($p = 0.200$ for both).

Older human organoids thus exhibited higher cross-unit synchrony and more regular firing — consistent with the maturation trajectory reported by Trujillo et al. [2019]. However, G1 recordings lasted 3 min and G2 recordings 10 min, creating an age–duration confound that cannot be fully decoupled at $n = 4$ per group. Replication with $n \geq 8$ and controlled recording duration is required.

3.4. Network Topology — Small-World Structure

Graph-theoretic analysis (STTC threshold 0.1, top 20 units) revealed heterogeneous topology across human organoids. Two subjects (HO6 and HO7, both P100D) exhibited strong small-world topology ($\sigma \approx 4.3$), while older organoids (>6 months) showed dense near-fully-connected

networks ($\sigma \approx 1.0$). This suggests a topological transition during organoid maturation from sparse small-world to hub-dominated architecture, consistent with developmental wiring principles [Watts and Strogatz, 1998].

Table 4: Graph theory metrics for human organoids (DANDI:001603).

Subject	Age	Density	Clustering	σ	Small-world
HO1–HO4	P6M–P7M	0.65–0.96	0.81–0.97	1.00–1.20	Weak
HO5	P100D	0.068	0.367	1.000	No
HO6	P100D	0.153	0.364	4.363	Strong
HO7	P100D	0.126	0.390	4.333	Strong
HO8	P100D	0.042	0.133	1.000	No

3.5. DANDI:001132 — SpikeInterface on HD-MEA Raw Data

Five subjects from DANDI:001132 (*Homo sapiens*, human hippocampal organoids) were analyzed. All files contained raw electrophysiology with 745–972 channels at 20,000 Hz. DC-corrected RMS channel selection (64/745–972 channels, RMS range 13–68 μ V) followed by mountainsort5 (scheme1) sorting was applied (Table 5).

Table 5: DANDI:001132 SpikeInterface sorting results. Qualification: SNR > 3.0, ISI violation < 10%, Hz > 0.05.

Subject	Channels	Duration (s)	Units	Qualified	Median Hz	Max STTC	Bursts
3C-1	972→64	41	15	9/15	1.76	0.196	3
4C-1*	745→64	180	2	0/2	19.76	−0.014	0
5C-1	942→64	51	6	4/6	1.11	0.058	0
6C-1	947→64	101	3	2/3	0.74	0.027	0
7D-7	907→64	81	4	3/4	1.10	0.028	0

*Excluded from comparative analyses (likely failed sorting; see Section 4.1).

3.6. DANDI:000774 — In Vitro MEA Validation

Five subjects from DANDI:000774 (sorted units, mouse in vitro MEA) were analyzed with per-unit analysis capped at 50 units. Recordings ranged from 3,261 to 10,926 s with high burst counts (514–1,708). Median firing rates ranged from 0.41 to 1.58 Hz with median STTC values 0.046–0.130, indicating moderate synchrony consistent with in vitro cortical network dynamics.

3.7. DANDI:001611 — Chronic Culture Scalability Test

One subject from DANDI:001611 (sub-24846, dissociated rat cortical culture, 950 sorted units, 261,569 s recording, 974,562 total spikes) was analyzed as a pipeline scalability test. Axon completed the full pipeline without modification in 16,225 s (≈ 4.5 hours), demonstrating that the pipeline handles large-scale chronic recordings. This subject is not included in organoid comparisons.

4. Discussion

Axon provides the first open-source pipeline specifically designed for NWB-formatted brain organoid MEA data with DANDI integration. Validation across four datasets demonstrated consistent performance across sorted-units and raw-signal data types, multiple species and preparation types, and recording durations from 41 s to 3 days.

The age-dependent synchrony finding ($p = 0.029$, $n = 4$ per group) is consistent with prior reports of emergent oscillations in maturing organoids [Trujillo et al., 2019]. The strong small-world

topology in HO6 and HO7 ($\sigma \approx 4.3$, both P100D) but not in HO5 and HO8 (also P100D, $\sigma = 1.0$) suggests that small-world organization may be a transient developmental feature. Age-matched longitudinal recordings with controlled duration are required for definitive interpretation.

The DC-offset correction step proved critical for HD-MEA channel selection. Without it, uint16-to-int16 converted recordings exhibit near-identical RMS ($\approx 32,000$ ADU), rendering channel selection uninformative. After mean subtraction, RMS variation (13–68 μV) reflects genuine differences in electrode activity.

4.1. Limitations

- **Age–duration confound (DANDI:001603):** G1 (3 min) and G2 (10 min) recordings differ in both age and duration. Replication at $n \geq 8$ with controlled duration is required.
- **Unit cap at 50 for per-unit analysis:** Prevents computational overload on high-unit-count preparations. Future work should implement vectorized ISI computation.
- **SpikeInterface run-to-run variation:** Stochastic clustering in mountainsort5 produces ± 1 unit variation per run. This should be quantified systematically.
- **4C-1 sorting failure:** Two units, 0 qualified, median Hz 19.76 for a 180-s recording — anomalous. Likely reflects unresolved multi-unit activity.
- **No isolation distance or L-ratio:** Waveform-based quality metrics beyond SNR are not yet computed automatically.

5. Conclusions

Axon provides a standardized, reproducible, open-source analytical baseline for brain organoid MEA electrophysiology. By integrating DANDI access, SpikeInterface spike sorting, and graph-theoretic analysis in a single command-line tool, Axon reduces the analytical barrier for groups generating or analyzing organoid MEA data. Validation across 29 subjects from four public datasets demonstrates consistent performance across diverse recording conditions and preparation types.

The pipeline is designed as a computational companion to organoid-oi v3 [Metin, 2026], a closed-loop organoid intelligence simulation system: Axon quantifies organoid dynamics from existing recordings, while organoid-oi provides the computational framework for future closed-loop hardware experiments.

Data and Code Availability

All datasets are publicly available on the DANDI Archive under CC-BY-4.0: DANDI:001603, DANDI:001132, DANDI:000774, DANDI:001611 (dandiarchive.org).

Axon source code: github.com/metin/axon.

organoid-oi v3: github.com/metin/organoid-oi.

No new biological data were generated for this study.

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